TED (21) – 2022 REVISION 2021

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

ELECTRICAL CIRCUITS AND MACHINES

MODEL QUESTION PAPER – SET-1

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	State the condition for maximum power transfer from source to load	M 1.01	U
2	Define form factor	M1.04	
3	In a three phase AC circuit, what is the sum of all the three generated voltages.	M1.04	
4	State the condition for resonance	M2.01	
5	Define Q- factor	M2.02	
6	What is the function of Commutator in DC machine	M3.01	
7	Explain the necessity of starters in DC motors	M3.04	
8	What is the advantage of salient poles in an alternator?	M4.01	
9	What is the major difference between a 3-phase induction motor and a single-phase induction motor?	M4.02	

PART B

II. Answer any Eight questions from the following

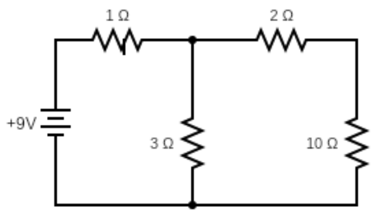
(8 x 3 = 24 Marks)

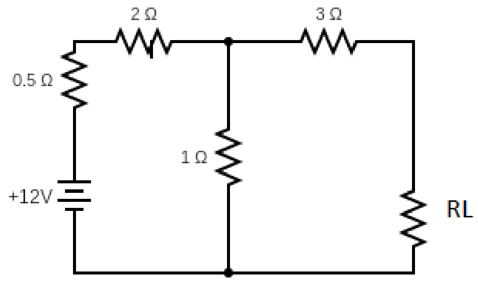
		Module Outcome	Cognitive level
1	State and explain Kirchoff‘s law	M 1.01	U
2	A parallel RLC circuit with 6-A trough the resistor, 8-A through the inductor, 5-A through the capacitor, calculate total line current. Is the circuit capacitive or inductive?	M1.04	
3	Define the terms apparent power, reactive power, and true power.	M1.04	
4	Derive an expression for energy stored in a capacitor	M2.03	

5	A 400V, 3-phase star connected alternator supplies a delta-connected load, each phase of which has a resistance of 30 ohm and inductive reactance 40 ohm. Calculate (a) the current supplied by the alternator and (b) the output power and the kVA of the alternator, neglecting losses in the line between the alternator and load	M2.03	
6	State the advantages of three phase system over single-phase system.	M2.03	
7	Derive EMF equation of a DC generator.	M3.02	
8	Explain armature control and field control of controlling speed of a DC motor	M3.05	
9	For an 8-pole induction motor supplied with power by a 6 pole Alternator at 1200 revolutions per minute, find the value of motor speed at slip of 3%.	M4.01	
10	How to make Single-Phase Induction Motor Self-Starting?	M4.03	

PART C

III. Answer all questions from the following (6 x 7 = 42 Marks)

		Module Outcome	Cognitive level
1	For the given circuit, determine the current flowing through 10 Ω resistor using Norton's theorem 	M1.03	
OR			
2	Determine the power factor of a RLC series circuit with R=5ohm, XL=8ohm and XC=12ohm.	M1.04	
3	Three capacitors 10 MFD ,25 MFD, 50 MFD are connected in series , find the equivalent capacitance and energy stored, when capacitors are connected across a 500V supply.	M1.04	
OR			

4	Find the value of RL at which maximum power is transferred to the load in the following circuit. Also, find the maximum power transferred 	M1.03	
5	Two wattmeters connected to a 3-phase motor indicate the total power input to be 12kW. The power factor is 0.6. Determine the readings of each wattmeter.	M2.04	
	OR		
6	Derive the q- factor for series resonance and parallel resonance	M2.02	
7	Explain the methods of speed control of DC shunt motor	M3.05	
	OR		
8	Draw the constructional details and Explain the working of DC generator	M3.02	
9	Explain the construction and working of three phase slip ring induction motor	M4.01	
	OR		
10	Explain the Open circuit and short circuit test in transformers.	M4.05	
11	Name the types of single-phase motors with respect to the method employed to make them self-starting? Describe the principle of operation of each type with suitable diagrams.	M4.03	
	OR		
12	Explain in detail the Starting of Slip-Ring Motors? What are the advantages of wound rotor motors over squirrel cage motors ?	M4.03	

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MODEL QUESTION PAPER – SET-2

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	State Thevenin's theorem	M 1.01	U
2	Distinguish between a Loop & Mesh of a circuit	M1.02	
3	What is a Resonant frequency?	M2.01	
4	Define Q- factor	M2.01	
5	Define back emf	M3.03	
6	Classify the DC Generators according to field excitation.	M3.02	
7	What are the advantages and disadvantages of three phase induction motor?	M4.01	
8	Describe the working principle of Alternator	M4.01	
9	Why single phase induction motor does not self starts?	M4.03	

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

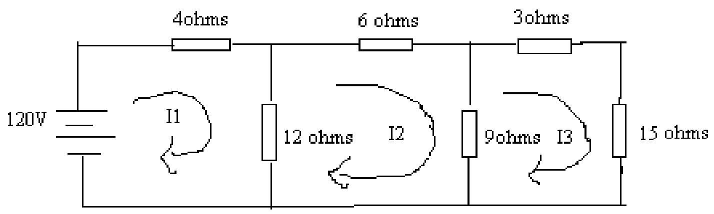
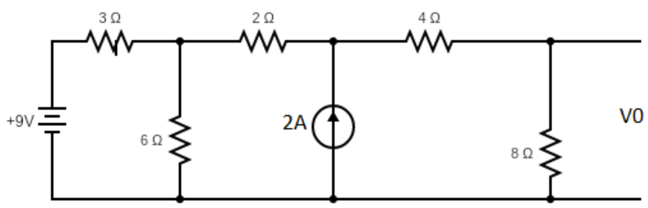
		Module Outcome	Cognitive level
1	State and explain Kirchoff's law	M 1.01	U
2	Using a parallel RC circuit which has a power supply of 100 – V, 60 Hz, and a current flow through the resister of is 10 amps and the current flow through the capacitor is 10 amps. Find a)The total current. b) True power c)Reactive power	M1.04	
3	Derive the relation between line voltage and phase voltage in a three phase delta connected circuit	M2.03	
4	Draw resonance curve for series resonance	M2.01	
5	A series L–R–C circuit has a sinusoidal input voltage of maximum value 12 V. If inductance, L = 20 mH, resistance, R	M2.01	

	= 80 Ω, and capacitance, C = 400 nF, determine the resonant frequency		
6	Three loads, each of resistance 30, are connected in star to a 415 V, 3-phase supply. Determine (a) the system phase voltage, (b) the phase current and (c) the line current	M2.03	
7	Draw the open circuit characteristics of separately excited DC generator and explain with neat sketch	M3.02	
8	Explain the necessity of starters in DC motors	M3.04	
9	Explain the working principle of capacitor start induction motor	M4.03	
10	Draw the equivalent circuit of single-phase transformer	M4.04	

PART C

III. Answer all questions from the following (6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Find the currents supplied by the batteries by using Mesh Analysis 	M1.02	
OR			
2	Find V0 using Thevenin theorem 	M1.03	
3	For RLC series circuit draw voltage triangle, power triangle and impedance triangle along with proper labelling's and equations for condition VL > VC.	M1.04	
OR			
4	Prove that the power in a R-L Series circuit is VI COSΦ	M1.04	
5	With neat diagram explain the measurement of three phase power by two wattmeter method	M2.04	
OR			

6	A series L–R–C circuit has a sinusoidal input voltage of maximum value 12 V. If inductance, L = 20 mH, resistance, R = 80 Ω, and capacitance, C = 400 nF, determine (a) the resonant frequency, (b) the value of the p.d. across the capacitor at the resonant frequency, (c) the frequency at which the p.d. across the capacitor is a maximum, and (d) the value of the maximum voltage across the capacitor.	M2.01	
7	Explain the working of 3-point starter for DC motor with neat sketch. Label theparts	M3.04	
OR			
8	Draw the open circuit characteristics of separately excited DC generator and explain with neat sketch	M3.02	
9	Explain the construction and working of three phase squirrel cage induction motor	M4.01	
OR			
10	Draw the phasor diagram of an ideal transformer on load at lagging and leading power factor	M4.04	
11	Describe the principal of operation of Split-Phase Induction Motor? What are the applications of split phase induction motor	M4.03	
OR			
12	Describe the Methods of Starting 3-Phase Induction Motors?	M4.02	